

Zheng Zhang

CONTACT INFORMATION	Institute of Mathematical Sciences ShanghaiTech University 393 Middle Huaxia Road Pudong, Shanghai 201210 China +86-21-20684807 zhangzheng@shanghaitech.edu.cn https://sites.google.com/view/zhengzhangmath
EDUCATION	Stony Brook University, USA, 2009-2014 Ph.D., Mathematics, December 2014 • Advisor: Radu Laza • Thesis: On geometric and motivic realizations of variations of Hodge structure over Hermitian symmetric domains. Zhejiang University, China, 2005-2009 B.S., Mathematics, June 2009 • Advisor: Shiu-chun Wong
EMPLOYMENT	• Assistant Professor, ShanghaiTech University, July 2020-present. • Visiting Assistant Professor, University of Colorado Boulder, August 2018-May 2020. • Visiting Assistant Professor, Texas A&M University, August 2015-July 2018.
RESEARCH INTERESTS	Algebraic geometry. In particular, Hodge theory and its geometric applications, special varieties (e.g. $K3$ surfaces, abelian varieties, cubic hypersurfaces, hyper-Kähler varieties), geometric and motivic realizations of variations of Hodge structure over Hermitian symmetric domains.
PUBLICATIONS	• On the image of the Prym map for double covers of genus-one curves branched at six points (with S. Casalaina-Martin, M. Lahoz and J.C. Naranjo), in preparation. • K-moduli, variations of GIT and period maps for cubic surface and cubic threefold pairs (with F. Si and C. Zhou), in preparation. • Moduli of Persson surfaces: compactification via KSBA stable pairs and a generic global Torelli type theorem (with H. Fang, B. Nguyen and X. Wu), arXiv:2605.17223, submitted. • Two results regarding the variation of K-moduli (with F. Si and C. Zhou), arXiv:2303.10963, submitted. • Infinitesimal Torelli problems for special Gushel-Mukai and related Fano threefolds: Hodge theoretical and categorical perspectives (with X. Lin and S. Zhang), arXiv:2507.06995, submitted. • Three approaches to a categorical Torelli theorem for cubic threefolds of non-Eckardt type via the equivariant Kuznetsov components (with S. Casalaina-Martin, X. Hu, X. Lin and S. Zhang), arXiv:2405.20554, submitted.

- The moduli space of cubic surface pairs via the intermediate Jacobians of Eckardt cubic threefolds, ZAG Handbook of Algebraic Geometry, Chapter 226, Chapman and Hall/CRC, 2025.
- Unimodal singularities and boundary divisors in the KSBA moduli of a class of Horikawa surfaces (with P. Gallardo, G. Pearlstein and L. Schaffler), Math. Nachr. 297 (2024), no. 2, 595–628.
- The moduli space of cubic threefolds with a non-Eckardt type involution via intermediate Jacobians (with S. Casalaina-Martin and L. Marquand), Int. Math. Res. Not. IMRN (2023), no. 18, 16104–16139.
- The moduli space of cubic surface pairs via the intermediate Jacobians of Eckardt cubic threefolds (with S. Casalaina-Martin), J. London Math. Soc. (2) 104 (2021), no. 1, 1–34.
- A generic global Torelli theorem for certain Horikawa surfaces (with G. Pearlstein), Algebr. Geom. 6 (2019), no. 2, 132–147.
- On motivic realizations of the canonical Hermitian variations of Hodge structure of Calabi-Yau type over type $D^{\mathbb{H}}$ domains, Canad. Math. Bull. 62 (2019), no. 1, 209–221.
- On the moduli space of pairs consisting of a cubic threefold and a hyperplane (with R. Laza and G. Pearlstein), Adv. Math. 340 (2018), 684–722.
- Compactifications of the moduli space of plane quartics and two lines (with P. Gallardo and J. Martinez-Garcia), Eur. J. Math. 4 (2018), 1000–1034.
- Classical period domains (with R. Laza), Recent advances in Hodge theory, 3–44, London Math. Soc. Lecture Note Ser., 427, Cambridge Univ. Press, Cambridge, 2016.
- A realization for a $\mathbb{Q}$ -Hermitian variation of Hodge structure of Calabi-Yau type with real multiplication, Math. Res. Lett. 22 (2015), no. 3, 967–982.

GRANTS

- Young Scientists Fund of the National Natural Science Foundation of China (NSFC), No. 12201406, 300,000 CNY, PI (2023–2025).

INVITED TALKS

- Weihai Workshop on Hodge Theory and Singularities, Shandong, July 2026.
- Algebraic Geometry Seminar, Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), December 2025.
- Southeastern Algebraic Geometry Symposium VIII, Southern University of Science and Technology (SUSTech), June 2025.
- Algebraic Geometry Seminar, Shanghai Jiaotong University, April 2024.
- Algebraic Geometry Seminar, IGP University of Science and Technology of China, November 2023.
- Geometry Seminar, Shantou University, October 2023.
- Algebraic Geometry Seminar, Xiamen University, June 2023 & March 2026.
- Moduli Spaces and Related Topics, YMSC Tsinghua University, May 2023.

- Conference on Algebraic Geometry and Related Topics in Nanjing, Nanjing University, May 2023.
- Algebraic Geometry Seminar, SCMS Fudan University, December 2022.
- Algebraic Geometry Seminar, Tongji University, November 2021, December 2022.
- Zoom Algebraic Geometry Seminar, online, July 2021.
- Joint Mathematics Meetings AMS Special Session on Algebraic Cycles in Arithmetic and Geometry, Denver, January 2020.
- Algebraic Geometry seminar, University of California at Santa Barbara, May 2019.
- Geometry Seminar, University of Texas at Austin, October 2017.
- Algebraic Geometry and Number Theory seminar, Rice University, September 2017.
- Hodge Theory, Moduli and Representation Theory, Stony Brook University, August 2017.
- The Georgia Algebraic Geometry Symposium (mini-talk), University of Georgia, March 2017.
- FRG Workshop, Washington University in St. Louis, January 2017.
- Geometry Seminar, Duke University, April 2016.
- FRG Workshop, Texas A&M University, January 2016.
- Geometry Seminar, Texas A&M University, April 2015 & September 2015 & November 2017.
- FRAGMENT Seminar, University of Colorado Boulder, February 2015 & September 2018.

RESEARCH ACTIVITIES

- Co-organizer of 2026 Pacific Rim Conference on Complex and Symplectic Geometry, August 2026.
- Co-organizer of Conference on Algebraic and Differential Geometry, June 2026.
- Co-organizer of Workshop on Calabi-Yau Geometry, October 2024.
- Organizer of ShanghaiTech-Tongji Algebraic Geometry Symposium, September 2022.
- Co-organizer of Algebra Seminar (ShanghaiTech University), 2020-present.
- Co-organizer of Algebraic Geometry Seminar (University of Colorado Boulder), 2018-2020.
- Co-organizer of Geometry Seminar (Texas A&M University), 2017-2018.
- [Collaborate@ICERM](#): Interplay among hypersurface singularities, Hodge theory and the moduli of hypersurfaces and their hyperplane sections (with P. Gallardo and J. Martinez-Garcia), May 2017.
- Postdoctoral member of [FRG \(Hodge theory, Moduli and Representation theory\)](#), 2014-2017.

TEACHING
EXPERIENCE

- Instructor, ShanghaiTech University
 - GEMA1001 - Higher Mathematics I Summer 2026
 - MATH 1123 - Complex Analysis Fall 2025
 - MATH 1224 - Abstract Algebra Spring 2025
 - MATH 2315 - Introduction to Riemann Surfaces Fall 2024
 - MATH 1224 - Abstract Algebra Spring 2024
 - MATH 2101 - Algebraic Topology Fall 2023
 - MATH 1224 - Abstract Algebra Spring 2023
 - MATH 1324 - Differential Topology Spring 2022
 - GEMA1010.01 - Mathematical Analysis II Spring 2021
 - GEMA1009.01 - Mathematical Analysis I Fall 2020
- Instructor, University of Colorado Boulder
 - MATH 2130-004 - Linear Algebra for Non-Math Majors Spring 2020
 - MATH 3450-001 - Introduction to Complex Variables Spring 2020
 - MATH 8174-001- Topics in Algebra ($K3$ surfaces) Fall 2019
 - MATH 3450-001 - Introduction to Complex Variables Spring 2019
 - MATH 2130-002 - Linear Algebra for Non-Math Majors Fall 2018
 - MATH 2130-003 - Linear Algebra for Non-Math Majors Fall 2018
- Instructor, Texas A&M University
 - MATH 304-501 - Linear Algebra Spring 2018
 - MATH 304-502 - Linear Algebra Spring 2018
 - MATH 308-512 - Differential Equations Spring 2017
 - MATH 308-521 - Differential Equations Spring 2017
 - MATH 308-503 - Differential Equations Fall 2016
 - MATH 308-506 - Differential Equations Spring 2016
 - MATH 308-517 - Differential Equations Spring 2016
 - MATH 308-506 - Differential Equations Fall 2015
- Instructor, Stony Brook University
 - MAT 514 - Analysis for Teachers II Summer 2015
 - MAT 123 - Introduction to Calculus Summer 2015
 - MAP 103 - Proficiency Algebra Spring 2013

SERVICE AND
OUTREACH

- Refereed for journals including Algebra & Number Theory, Épijournal Géom. Algébrique, Exp. Math., Rev. Mat. Iberoam., SIGMA Symmetry Integrability Geom. Methods Appl. and Trans. Amer. Math. Soc.
- AMS Mathematical Reviews and Zentralblatt MATH.
- Serve on Undergraduate Recruitment Committee and Graduate Studies Committee (ShanghaiTech University).
- Serve on Yu Wang's Qualifying Exam Committee (University of Colorado Boulder).
- Undergraduate thesis advisor: Zhiyao Pu and Yancheng Zhu (2026); Yiyang Gong (2025).
- Tutor for undergraduate students (ShanghaiTech University): Sichen Du and Chenyu Xu (2025-2026); Zilin He (2024-2025); Yuxin Cui, Lecong Ding and Hongyao Li (2023-2024); Yu Chen and Yifan Wang (2022-2023); Junru Li, Guanqiu Wu and Beining Xu (2021-2022); Yitong Liu and Yipeng Zhang (2020-2021).

REFERENCES

- Sebastian Casalaina-Martin, University of Colorado Boulder, casa@math.colorado.edu
- Phillip Griffiths, Institute for Advanced Study, pg@math.ias.edu
- Radu Laza, Stony Brook University, radu.laza@stonybrook.edu (Ph.D. advisor)
- Gregory Pearlstein, Texas A&M University, gpearl@math.tamu.edu